Online Resource 1

Catatonia recovery separates two intervention axes that constrain electroconvulsive therapy

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1 Zero-evidence-rate control for the escape-time exponent

The escape time above the critical coupling diverges as a power of the excess with exponent one half (Sect. 7 of the main text), and two separate effects push a fitted exponent away from that value. The first is pre-asymptotic curvature: at any finite excess the fit still contains the approach to the asymptote, so it exceeds one half over a wide range and falls towards one half as the excess is reduced. The second is feedback through the evidence coordinate: while the system waits on the branch, Q rises, a(Q) falls and the critical coupling moves, so the excess against which the fit is taken is not the excess the system experiences. The first effect is present in the capacity subsystem, which has no evidence channel; the second exists only at a positive evidence rate.

The control separates them by a single change. The full five-dimensional sweep is repeated with ε set to zero and everything else, including the six evidence levels, the starting point at half the critical coupling for the prevailing threshold, the excess grid and the solver settings, held fixed. With the evidence channel off the capacity block depends on the evidence level only through the constant a, and the excess is measured in units of the corresponding κ_c; the asymptotic exponent is then one half at every level, but an exponent fitted over a finite range of excess may still differ between levels, because the scaled first-stage equation retains a dependence on a (below).

Table S1 Exponents of the full system with the evidence channel on and off. Each entry is the negative slope of log escape time against log excess, fitted over the whole grid (0.005 to 0.8) and restricted to excesses at or below 0.05

Q_0	$a(Q_0)$	ε = 0.02, full	ε = 0.02, ≤ 0.05	ε = 0, full	ε = 0, ≤ 0.05
1.00	0.1500	0.570	0.532	0.570	0.532
0.99	0.1590	0.530	0.456	0.570	0.532
0.90	0.2355	0.501	0.432	0.571	0.532
0.70	0.3795	0.484	0.401	0.572	0.531
0.50	0.4875	0.478	0.386	0.574	0.529
0.00	0.6000	0.516	0.430	0.577	0.527

With the channel off the six levels agree to within 0.007 in the full-range exponent (0.570 to 0.577) and to within 0.005 in the small-excess exponent (0.527 to 0.532), across a threshold range of a factor of four, against a spread of 0.146 in the same column with the channel on. Switching the channel off therefore removes most of the spread between levels; the small spread that remains is not analysed further, and since the first-stage equation in scaled variables, $dy/d\tau = (1 - ay)(y^2 - y + K)$ with $r_1 = ay$, $\kappa = a^2K$ and $\tau = at$, retains a dependence on a, a finite-range exponent is not expected to be identical across thresholds even where the asymptotic exponent is one half in each case. At $Q_0 = 1$ the two sweeps agree to every digit reported, because $z > 0$ on the branch makes the only inflow term proportional to $1 - Q$, which vanishes there, so the evidence coordinate is stationary and ε cannot act.

Continuing the control to smaller excess shows the approach to the asymptote over the range computed. At $Q_0 = 0.5$ the local slope between consecutive excesses is 0.512 at 3.2×10^{-3}, 0.508 at 1.4×10^{-3}, 0.505 at 7.1×10^{-4} and 0.504 at 3.2×10^{-4}; at $Q_0 = 0$ the same sequence is 0.510, 0.507, 0.505 and 0.503. Both approach one half from above at the two thresholds examined, which differ by a factor of 1.23.

One alternative explanation is excluded arithmetically rather than numerically. Every sweep starts on the exact partial branch at $\kappa = \kappa_c/2$, and the first stage there is $r_1 = a(1 - 1/\sqrt{2})/2$, while the branch remnant at κ_c is $r_1 = a/2$. The ratio is $1 - 1/\sqrt{2} = 0.2929$ at every threshold, since a cancels. The starting point is therefore the same fraction of the way to the fold at every evidence level, so a difference in the normalised initial position does not account for a spread across levels; a difference in the dynamics through the threshold a is not excluded by this argument.

2 Waiting times to the annihilation locus

On the intermediate band (Proposition 7 of the main text) the time to reach $Q*(\kappa)$ is set by ε $C(r)$ and by the selector scale. In the full-system integrations it ranges from 8.8×10^5 time units at $\kappa = 0.08$ with the narrow selector ($s = 0.01$) to 3.4×10^8 at the same coupling with the wide one ($s = 1$), a factor of 380, and to 4.9×10^7 at $\kappa = 0.05$ with the narrow selector. Below $a_1^2/4$ the locus does not exist and no waiting time is defined. The existence of the band, its two boundaries $a_1^2/4$ and $a_0^2/4$, the locus $Q*(\kappa)$ and the ordering of the fold and the escape follow from the geometry and are reported as results in the main text; the absolute waiting time is not a prediction, since it inherits the evidence rate and the regularisation of Saito (2026a) and varies by more than two orders of magnitude between the two selector scales. No clinical timescale is claimed from it.

Reference

Saito H (2026a) Evidence-coupled bistability converts dependency structure into a recovery geometry. Research Square, preprint. https://www.researchsquare.com/article/rs-10620384